FINAL CAPSTONE PROJECT REPORT

Stack Overflow in the AI Era

Question Resolution, Activity Forecasting and Text Analysis

Submitted for the course

23CSE452 BUSINESS ANALYTICS

Team 7

Table 1. Project team

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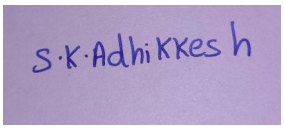
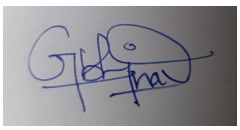
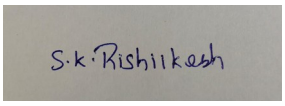
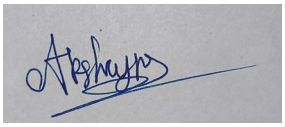
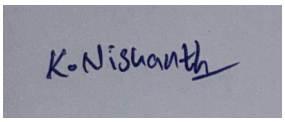
November 2026

Declaration and Acknowledgement

Declaration

We, the members of Team 7, declare that this report presents our capstone project, Stack Overflow in the AI Era, carried out for 23CSE452 Business Analytics under the guidance of Ms. Rohini S. The report describes our data collection, analytical implementation, evaluation and interpretation. External datasets, methods, studies and software are acknowledged through references. We accept responsibility for the accuracy of the recorded work and for explaining the methods and their limitations.

Table 2. Declaration by the project team

Name	Register number	Signature
Adhikkesh S K	CB.SC.U4CSE23101	
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Acknowledgement

We thank Ms. Rohini S for her guidance and feedback, and the faculty of the Amrita School of Computing for this project opportunity. We acknowledge Stack Exchange for the public API and source records, and the open-source maintainers whose software supported our analysis.

Abstract

Developer communities need evidence to allocate expert attention. This project studies Stack Overflow activity and an accepted-answer posting proxy using independently collected Stack Exchange API data from January 2019 to August 2026. Cleaning produced 302,026 unique questions with 39 variables and a separate table of 5,600 weekly observations. Exploratory analysis guided structural, temporal, account-age and title-text features. Logistic regression, random forest and XGBoost were compared chronologically; the selected original model achieved locked-test average precision of 0.4315. A calibrated release achieved 0.4892 on a separate retrospective sample of 389 August questions. Time-series analysis examined seasonal patterns, structural changes and forecasts. SARIMAX achieved 55.9% mean skill over naive at 26 weeks, although original 95% intervals covered only 52.0% of outcomes. Text mining selected 15 NMF topics using coherence, diversity and stability. A working Streamlit application combines historical evidence, individual predictions and business scenarios. The findings support monitored review and documentation pilots. Sampling, snapshot labels, provisional topic interpretations and uncertain forecast bands limit operational claims.

Keywords: Stack Overflow; chronological classification; SARIMAX; text mining; NMF; developer support.

Abbreviations

Table 3. Abbreviations used in the report

Term	Meaning
API	Application programming interface
AP (PR-AUC)	Average precision; a summary of precision-recall ranking
TF-IDF / SVD	Term frequency–inverse document frequency / singular value decomposition
STL / ITS	Seasonal-trend decomposition using LOESS / interrupted time series
SARIMAX	Seasonal autoregressive integrated moving average with exogenous inputs
NMF / LDA	Non-negative matrix factorisation / latent Dirichlet allocation
NPMI	Normalised pointwise mutual information
MASE / sMAPE	Mean absolute scaled error / symmetric mean absolute percentage error
HAC / SHAP	Heteroskedasticity and autocorrelation consistent / Shapley additive explanations

Executive Summary

The business problem is to allocate limited developer-support effort while understanding how the volume and composition of questions are changing. Our project combines individual question records with separately collected weekly totals. This lets us examine both the likelihood of a recorded resolution outcome and the broader demand for support.

The collected question sample shows a weighted resolution-proxy rate of about 44.1% before the transition and 35.3% afterwards. Weekly activity also declines across the study period. Our selected sitewide forecast projects approximately 134 questions per week by 23 February 2027, compared with 215 in the final observed week. The original forecast bands are too narrow in backtesting, so the trajectory is a planning scenario rather than a reliable staffing commitment.

The classifier provides a modest improvement in prioritisation. In the separate August evaluation, reviewing the 78 lowest-scored questions identifies 58 proxy-negative cases; random selection at the same capacity would identify about 50.93 in expectation. These cases are candidates for inspection. A negative proxy does not establish that a question lacks a useful answer, and finding a case does not establish that expert intervention will resolve it.

Text analysis adds context to the volume results. Docker, Android and API themes gain relative share in the collected corpus, while array and collection themes lose share. Questions also contain more prose words and code lines at the weighted median. These are changes in composition and structure, not independent measures of difficulty. General sentiment scoring agrees poorly with the existing human reference and should not control access to support.

We recommend a small comparison between model-ranked and randomly selected review queues, targeted documentation experiments for interpretable themes, and monitoring of actual forecast outcomes. Under explicitly assumed costs and intervention success, the base routing scenario yields about INR 851 incremental net value per 1,000 questions. The lower-success scenario is slightly negative. Business benefit must therefore be measured in a controlled pilot rather than inferred from model scores alone.

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1 Introduction Objectives and Literature

1.1 Business problem and analytical questions

We examine question-resolution outcomes, activity trends and technical themes across 13 technologies and a separate sitewide series, connecting three analyses to practical recommendations.

Table 6. Objectives and success checks

Objective	KPI and comparison	Success check
Prioritise expert review	Capture of proxy-negative cases at 20% capacity; random capture ≈20%	Capture above random, followed by a measured pilot
Plan weekly support	26-week error against naive; nominal 95% interval coverage	Positive point skill; coverage assessed separately
Identify support themes	Weighted topic share; NPMI, diversity and seed stability	Diversity ≥0.7 and stability ≥0.6; interpretable themes

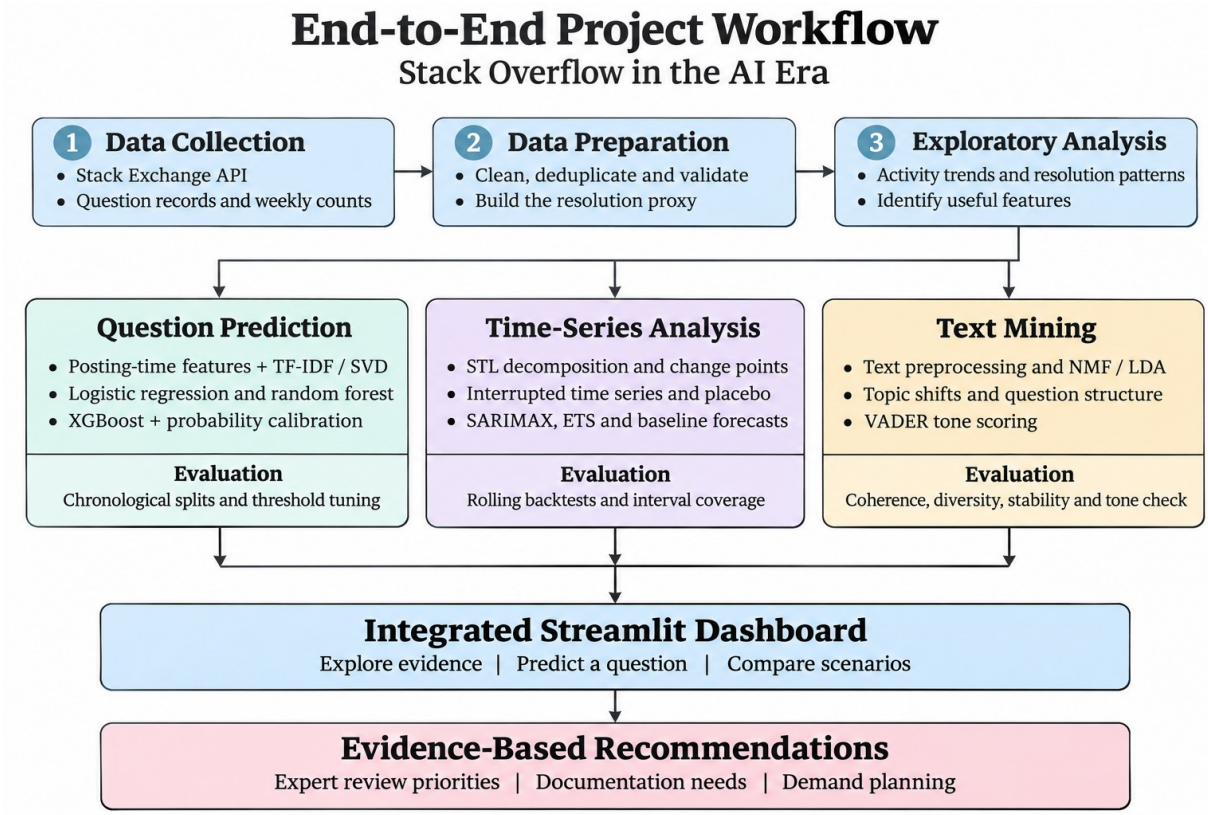


Figure 1. End-to-end analytical workflow. Source: project implementation.

1.2 Related work and contribution

Del Rio-Chanona et al. [1] estimate a 25% relative activity reduction across platforms after a late-2022 release; their counterfactual design supports stronger attribution than our time break. Kabir et al. [2] find incorrect information in 52% of 517 generated programming answers. Their answer-quality outcome differs from our resolution proxy. Our contribution integrates three analyses of a later period in one application.

2 Dataset Collection and Preparation

2.1 Source coverage and collection

We collected records directly through the Stack Exchange API v2.3 [3]. The study window is January 2019–August 2026, with the cleaned-data collection date recorded as 3 October 2026. The study tags are python, javascript, java, sql, pandas, reactjs, android, excel, r, docker, powerbi, fastapi and openai-api.

Before 2025, the design generally sampled one weekday and one weekend day per tag-month; small tag-months were collected exhaustively. Later periods attempted complete collection for the configured tags. Inverse-probability weights of 1–23 retain the known sampling design. Pagination, quota limits and temporary failures were handled through batching, caching, retries and resumable collection. Answer and user metadata were fetched separately for timing and account-age features.

Table 7. Collection and final-data audit

Item	Recorded result
Raw question records	309,179
Duplicate records removed	7,147
Other invalid records removed	6
Final unique questions / variables	302,026 / 39
Missing account age retained with flag	6,319
Surviving truncated-prose records	302
Weekly totals	14 series × 400 weeks = 5,600; no missing weeks
Final structural validation	All eight checks passed

2.2 Representative final records

Table 8. Five records from the final dataset

Question ID	Posted	Weight	Code blocks	Proxy / any answer
54160622	2019-01-12	8	3	1 / 1
54873725	2019-02-25	20	1	0 / 0
57098741	2019-07-18	23	4	1 / 1
57590610	2019-08-21	22	1	0 / 1
58675228	2019-11-02	9	1	1 / 1

Question links follow https://stackoverflow.com/q/{ID}. Proxy and any-answer indicators use the seven-day posting window. Full field definitions: docs/data_dictionary.md.

2 Dataset Quality and Measurement

2.3 Cleaning and field preparation

Tag collections were merged and deduplicated by question ID, timestamps were converted to UTC, and HTML was separated into prose and code. Missing account age was retained with a flag. In the revised model, those flagged values are restored to missing and the median is learned within training. Scaling, title TF-IDF and SVD are likewise fitted within the model pipeline rather than on the evaluation records.

Table 9. Field preparation and analytical use

Field group	Type and preparation	Use
created_at; tags	UTC timestamp; multiple technology indicators	EDA, temporal split, text periods
title; plain_body; code_text	Text; HTML removed and code separated	Title prediction; prose mining; structure
code blocks; code ratio	Count; code characters / total characters	Prediction and complexity
account age; missing flag	Days at posting; fold-fitted median imputation	Prediction and subgroup checks
target_resolved_7d	Binary accepted-answer posting proxy	Prediction and outcome summaries
sampling_weight; weekly count	Survey weight; separately collected integer total	Weighted rates; forecasting

2.4 Outcome definition and validity

The primary target is positive when the answer marked accepted in the collected snapshot was posted within 168 hours of the question. Acceptance-event time is unavailable. The secondary target records any answer posted within that window. A negative primary label can include useful answers that were never accepted, so the model identifies review candidates rather than confirmed support failures.

2.5 Bias privacy and permitted use

Known sampling weights address unequal inclusion, but they do not remove differences between sampled days, omitted technologies or inaccessible records. Tags overlap and must not be added as independent populations. Snapshot text can contain later edits. Public exports minimise personal information by omitting bodies and user hashes; identifiers in analytical data are pseudonymised rather than claimed anonymous. Source content retains Stack Overflow attribution and links under the recorded CC BY-SA 4.0 terms [3]. Applicable institutional data-handling requirements govern access and retention.

3 Exploratory Analysis and Visualization

3.1 Activity and resolution over time

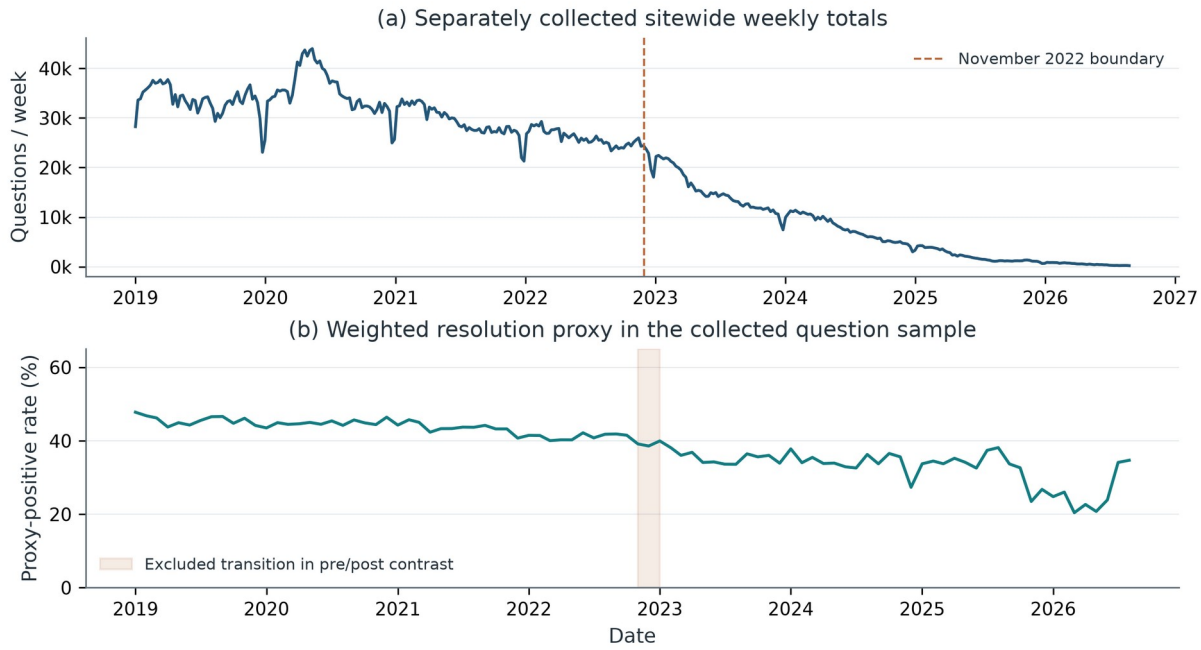


Figure 2. Weekly sitewide activity and monthly weighted resolution proxy. Source: weekly_counts.csv and monthly_rates_masked.csv. Tool: Matplotlib.

The weekly totals show long-term contraction, including decline before November 2022. The lower panel measures a different population: outcomes in the collected question sample, adjusted for the known sampling weights. The fixed comparison uses January 2019–October 2022 and January 2023–August 2026, excluding the transition. Its weighted proxy-positive rate falls from about 44.1% to 35.3%, a difference of approximately 8.7 percentage points using unrounded values.

3.2 Implications for modelling

Recorded EDA shows small individual relationships: categorical associations have Cramér's V below 0.11 and absolute numeric Spearman correlations are below 0.12. Large samples can make small effects statistically detectable. These findings support combining several signals while retaining realistic performance expectations. Changing volume, topic mix and question structure also make chronological evaluation more appropriate than a random split.

Reproduction: notebooks/review1_main.ipynb; weighted monthly tables; _build/generate_figures.py in this report folder. Months with fewer than 30 sampled questions are omitted from the displayed rate series.

4 Feature Engineering and Prediction

4.1 Predictors and preprocessing

The model uses question information intended to be available at posting: title and body length, code blocks, code ratio, error-text indicators, question word, tags, account age and calendar fields. Title unigrams and bigrams are represented with TF-IDF, capped at 5,000 terms, then compressed to 100 SVD components [5]. Categorical values are encoded, numerical values are imputed and scaled, and binary indicators are retained.

Table 10. Feature groups and rationale

Group	Examples	Reason for inclusion
Structure	Title/body length; code blocks; code ratio	Represent how the question is presented
Title text	TF-IDF with 100 SVD components	Capture recurring technical wording
Account and calendar	Account age; missing/new account; hour/day/month	Represent posting context
Technology	13 tag indicators; other tag count	Capture ecosystem differences
Past activity	Four-week volume; change from preceding four weeks	Represent recent support demand

4.2 Leakage controls

Votes, views, answer counts, outcome delays and other outcome-revealing fields are excluded. Sampling weights affect model fitting but are not predictor columns. Recent-volume features use only complete weeks before the posting week; the recent four weeks are compared with the four weeks before them. Questions without the required eight-week history are excluded from this modelling input.

Preprocessing is refitted within training folds. The revised release also separates fitting, calibration and threshold selection with seven-day gaps at relevant boundaries. Question IDs do not overlap across periods. Returning askers can occur in later periods, and original posting-time text revisions are unavailable; these limits remain part of the evaluation scope.

4.3 Algorithms

Logistic regression provides a simple reference. Random forest combines randomized decision trees, while XGBoost builds trees sequentially to reduce remaining prediction error [4]. Class weighting addresses imbalance. Model selection uses average precision, which evaluates ranking of proxy-positive cases across thresholds; it is not an accuracy percentage.

4 Model Selection and Evaluation Protocol

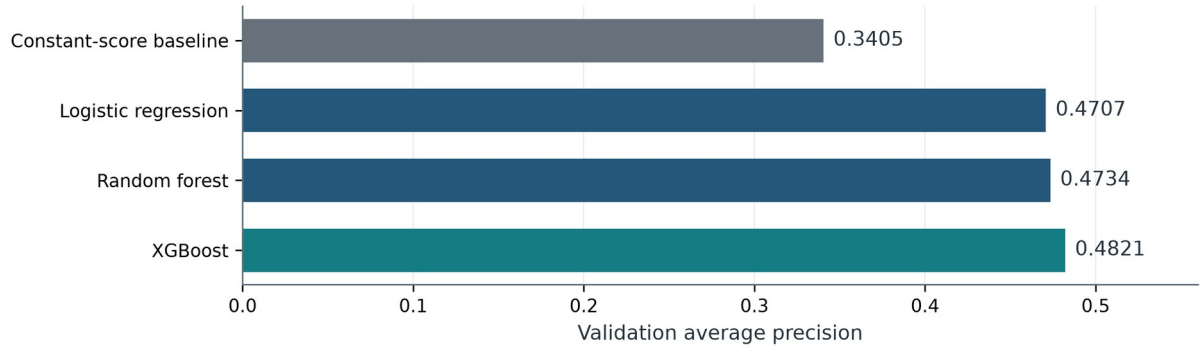


Figure 3. Original models on the same January–June 2025 validation set, n = 28,132. Source: preserved Review 1 model results. Tool: Matplotlib.

XGBoost was selected at validation average precision 0.4821, slightly above random forest at 0.4734 and logistic regression at 0.4707. Original tuning used eight random configurations, three chronological folds and a 50,000-row tuning subset with seed 42. Early stopping determined the retained tree count. The original model then underwent a locked evaluation on 16,695 July 2025–July 2026 questions.

Table 11. Separate model releases and chronological stages

Release / stage	Period	Questions
Original training	Before January 2025	248,294
Original validation	January–June 2025	28,132
Original locked test	July 2025–July 2026	16,695
Revised training	26 Feb 2019–24 Dec 2024	248,137
Revised calibration	1 Jan–31 Mar 2025	17,643
Revised threshold selection	8 Apr–23 Jun 2025	8,804
Revised retrospective evaluation	August 2026	389

The revised release freezes the selected XGBoost settings: depth 5, learning rate 0.0294479 and 628 trees. A sigmoid mapping calibrates raw probabilities using the separate calibration period. F1 selects the final threshold, 0.2971, on the later threshold period. No August outcomes are used in those fitting stages. August had already been examined in Unit 3, so this release has additional retrospective evidence rather than a blind prospective test.

4 Results Interpretation and Subgroups

Table 12. Classifier results on their own evaluation periods

Evaluation	AP	ROC-AUC	Precision	Recall	F1
Original locked test, n=16,695	0.4315	0.6548	0.3626	0.7712	0.4933
Revised August, n=389	0.4892	0.6198	0.3958	0.7037	0.5067

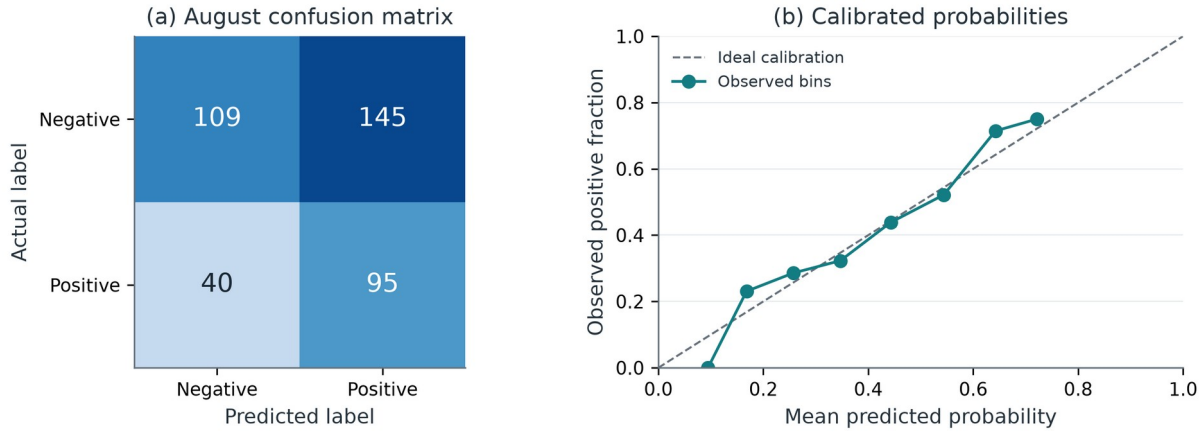


Figure 4. Revised August confusion matrix at threshold 0.2971 and calibration bins. Source: final_validation/confusion.csv and calibration.csv. Tool: Matplotlib.

The revised classifier detects 95 of 135 proxy-positive cases, but also predicts positive for 145 proxy-negative cases. Calibration reduces Brier probability error from 0.2218 to 0.2155. Different test periods prevent treating the two AP values as a controlled improvement comparison. August positive prevalence is 0.3470; its sample is small.

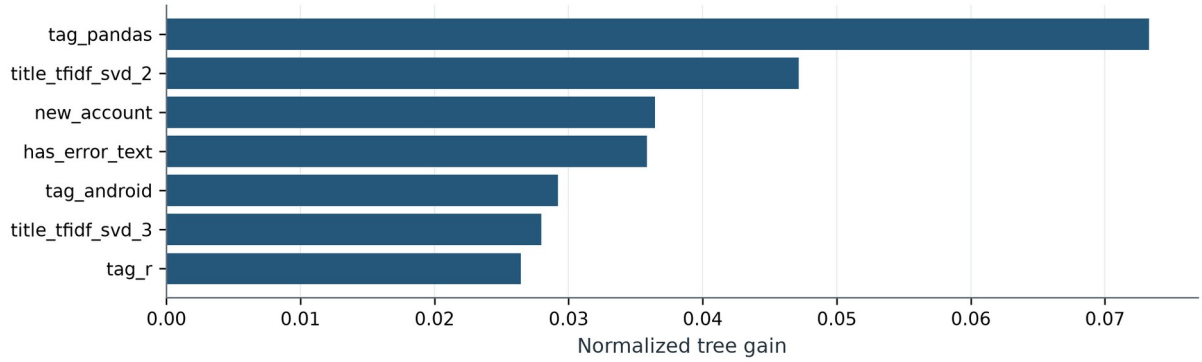


Figure 5. Largest global feature gains in the revised XGBoost model. Source: final_validation/feature_gain.csv. Tool: Matplotlib.

Tag membership, latent title features and new-account status contribute to model splits. Gain and local tree SHAP describe model behaviour rather than causal effects of editing a question. New-account recall is 0.60 (71 records), versus 0.722 for other accounts (318 records). Only nine records have missing age, making that subgroup estimate unstable. These differences support monitoring rather than a fairness verdict.

5 Advanced Analytics Time Series

5.1 Method 1 Preparation and historical patterns

Time-series analysis uses the separate weekly-count table: 400 Tuesday-start weeks for ALL and 13 technologies, with no missing weeks. Counts are transformed with log(1 + count), which accommodates zero and reduces scale differences. Robust STL separates trend, annual seasonality and remainder using period 52 [6].

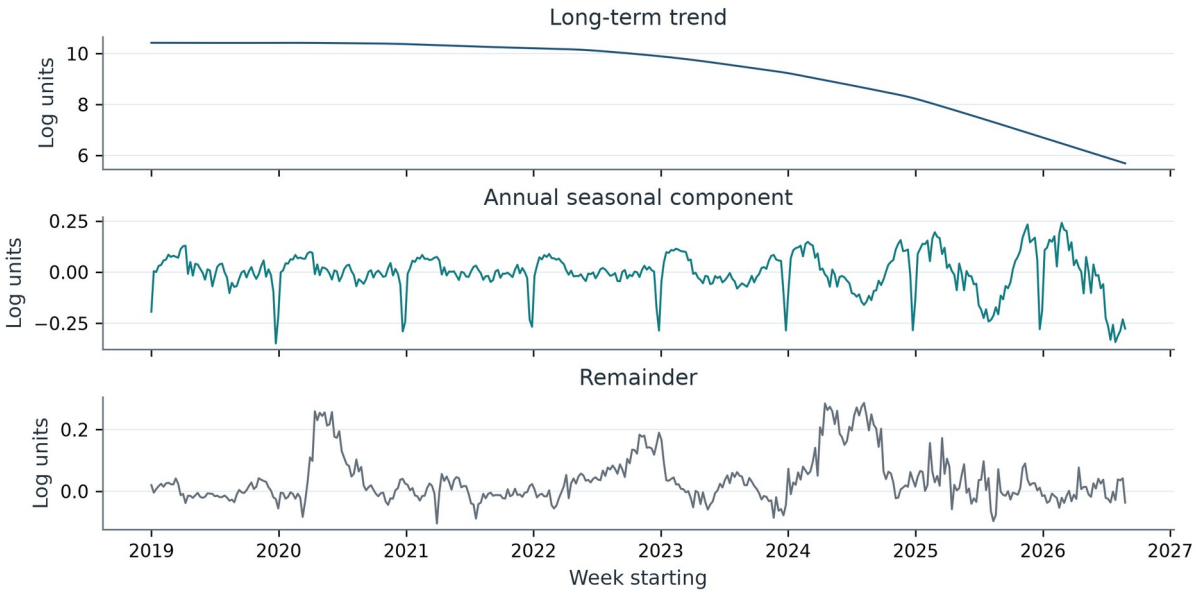


Figure 6. STL components for the ALL log-count series. Source: ts/stl_components.csv. Tool: Matplotlib.

The trend component captures sustained contraction, while seasonal and remainder components describe variation around it. ADF and KPSS tests indicate nonstationarity in levels across all 14 series. First-difference conclusions conflict in 12 series and support stationarity in two. The forecasting model therefore uses d=1 as a stated modelling choice, without claiming that every transformed series satisfies all assumptions.

PELT detects mean shifts in log counts with a minimum 26-week segment and a fixed penalty. The recorded run finds 134 breaks, six within four weeks of the four listed events. These proximity counts are descriptive and do not constitute a causal event test. Implementation: src/ts_patterns.py.

5 Structural Changes and Timing Comparison

5.1.1 Interrupted time series

We fit a segmented regression around the Tuesday-start week containing 30 November 2022. The design includes the earlier trend, an immediate level change, a later slope change and two annual Fourier pairs. Newey–West HAC uncertainty with four lags allows for correlated weekly errors. A November 2021 placebo repeats the comparison at an earlier boundary.

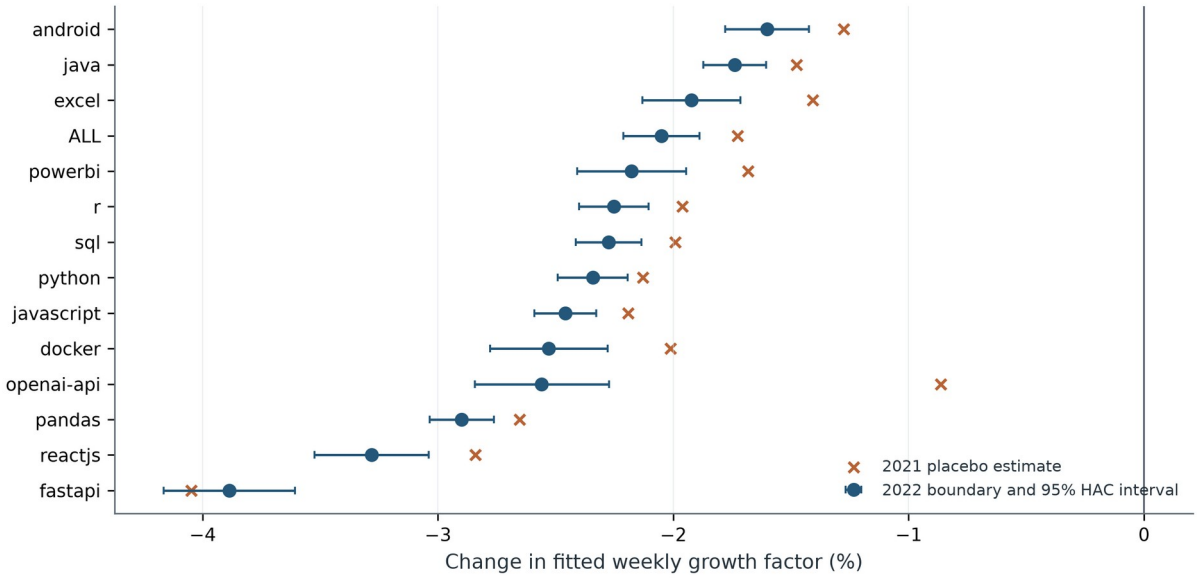


Figure 7. Estimated changes in fitted weekly growth factors and 95% HAC intervals; crosses show the earlier placebo estimates. Source: ts/its_results.csv. Tool: Matplotlib.

All 14 fitted slope adjustments are negative, ranging from approximately −3.89% to −1.60%. For ALL, the adjustment is −2.05%, with an interval of about −2.21% to −1.89%. These are transformations of the slope-change coefficient, exp(beta3) − 1, and describe a change relative to the earlier fitted growth pattern. They do not mean that the observed count falls by exactly 2.05% every week.

In 13 series, the actual-boundary adjustment has a larger magnitude than its placebo counterpart. The comparison strengthens the timing evidence, but the placebo also shows changes. Prior decline, platform changes and other events can contribute. We therefore report a temporal association, not a causal substitution estimate.

5 Forecast Backtesting and Error

5.1.2 Fair comparison of forecasting methods

Naive repeats the latest value; seasonal naive repeats the corresponding annual observation; drift extends the average training-history change. ETS smooths level and a damped trend. SARIMAX uses first differencing, AR/MA orders searched from 0–2, and known boundary and annual Fourier inputs [6,7]. Orders are chosen on earlier training data using AIC; later validation selects the model family. Expanding-origin tests use matched 1-, 4-, 13- and 26-week forecast cases.

Table 13. Mean matched 26-week performance across 14 series

Model	MASE	sMAPE %	Skill %	95% coverage %
Naive	0.723	88.5	0.0	70.8
Seasonal naive	2.098	138.3	−194.3	19.5
Drift	0.564	74.6	22.2	83.3
ETS	0.584	77.1	21.2	59.7
SARIMAX	0.324	55.7	55.9	52.0

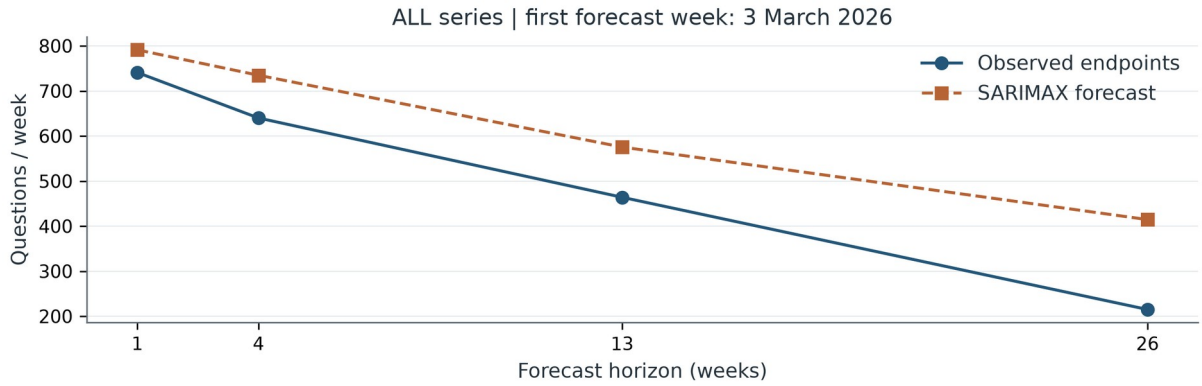


Figure 8. One historical forecast compared with known outcomes. Points are the stored horizons, not a complete weekly trajectory. Source: ts/backtest_predictions.csv. Tool: Matplotlib.

MASE scales count error by training-period one-step changes; sMAPE is already a percentage. Skill is computed against matched naive errors within each series and then averaged. SARIMAX has the lowest mean 26-week point error, but the plotted case also shows substantial overprediction. The ALL residual Ljung–Box p-values are 0.565 and 0.481 at lags 10 and 20; these checks do not establish calibrated prediction bands.

5 Forecast Scenario and Uncertainty

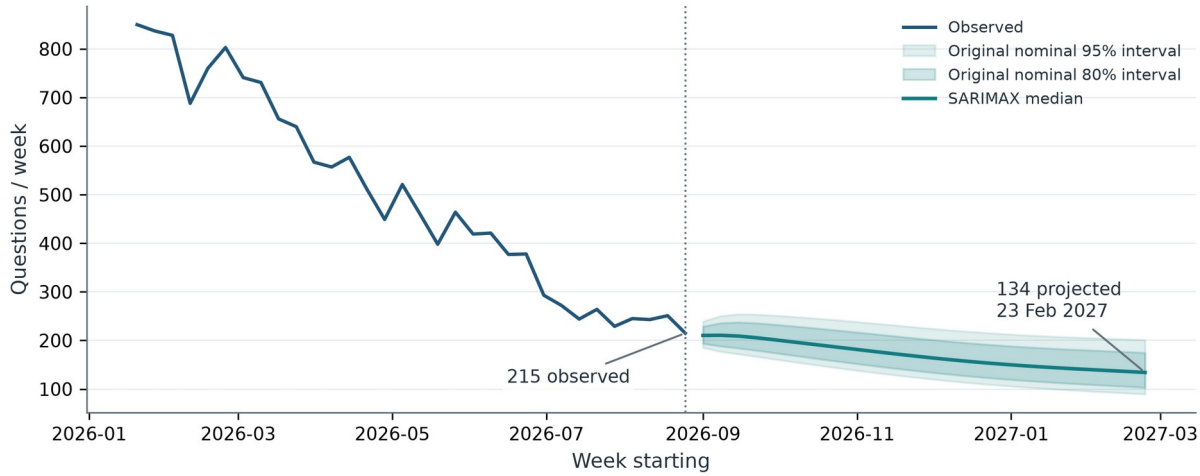


Figure 9. Selected ALL forecast with original uncertainty bands. Source: weekly_counts.csv and ts/forecasts_26w.csv. Tool: Matplotlib.

The selected sitewide SARIMAX median falls from 215 observed weekly questions to 134.16 by 23 February 2027, a projected decline of 37.6%. Back-transformation from the log scale gives a median rather than an arithmetic mean. Six of 13 technology tags meet the steep-decline rule below −40%; five are in the −40% to −10% declining range. Two have a zero last-week baseline, so their percentage changes are undefined.

Table 14. Illustrative technology forecast endpoints

Series	Selected model	Last week	Week 26	Change
ALL	SARIMAX	215	134.16	−37.6%
android	SARIMAX	10	6.01	−39.9%
docker	Drift	1	0.43	−56.9%

5.1.3 Reliability of the prediction intervals

Original nominal 95% SARIMAX bands cover only 52.0% of outcomes in the matched 26-week comparison. A separate diagnostic estimates interval inflation from outcomes before January 2026 and evaluates later origins. For pooled SARIMAX 26-week endpoints, coverage rises from 58.5% to 100% across 41 endpoints after widening by factor 3.594. The samples differ from the original summary. Overlapping horizons, few endpoints and changing errors prevent a guarantee of future coverage. The original future bands remain visible rather than being silently replaced.

5 Advanced Analytics Text Mining

5.2 Method 2 Corpus and topic selection

Text mining examines which technical themes occur in the questions and how their relative shares change. The corpus combines titles and prose while excluding code. Technology tokens are protected, URLs and paths are normalised, boilerplate is removed, and spaCy supplies lemmas [8]. Sampling weights and dates are retained for later summaries.

NMF uses TF-IDF and LDA uses word counts over the same vocabulary [5]. We compare six topic counts (10, 15, 20, 25, 30 and 40) and three seeds for each family: 36 actual fits on 75,814 questions, with a separate 20,000-document coherence reference. The reference is used for model selection, not treated as an untouched final test.

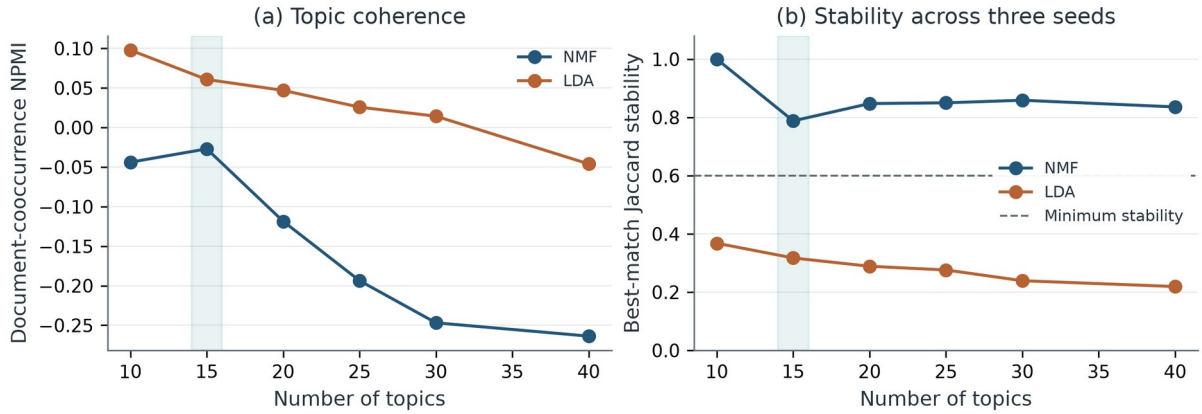


Figure 10. Measured coherence and seed stability across topic counts; shading identifies k=15. Source: text/topic_model_selection.csv. Tool: Matplotlib.

Table 15. Selected candidates from the measured grid

Candidate	NPMI	Diversity	Stability	Meets rule
NMF 10	−0.044	0.800	1.000	Yes
NMF 15	−0.027	0.735	0.788	Yes
LDA 10	0.098	0.715	0.368	No
LDA 15	0.061	0.726	0.317	No

Selection maximises document-cooccurrence NPMI among models with diversity ≥0.7 and stability ≥0.6. NMF 15 is selected. LDA's higher coherence does not overcome its weak stability. Negative NMF coherence and mixed topics remain interpretation limits. Names are automatic descriptions of leading words, not independently established categories. Implementation: src/text_prep.py and src/topics.py.

5 Topic Shifts Tone and Structure

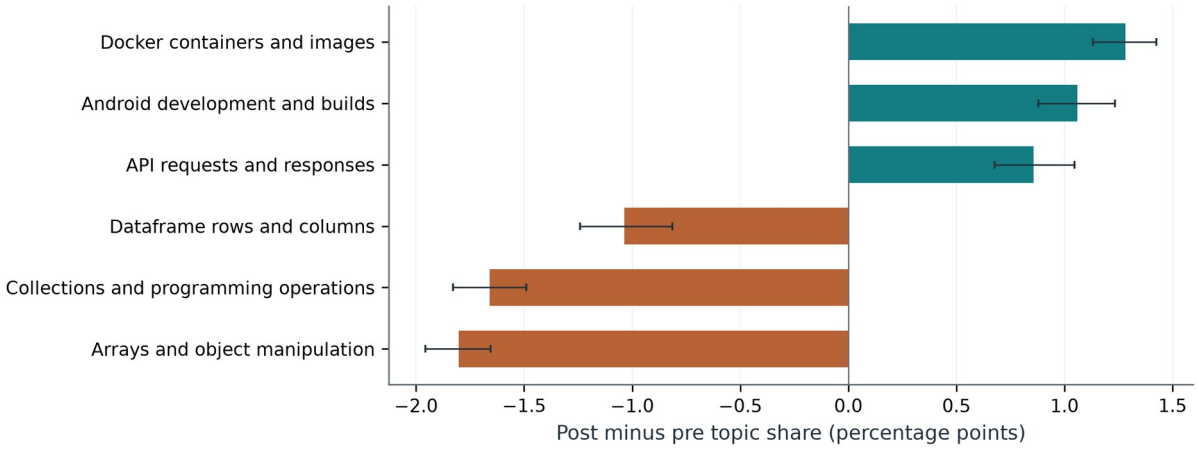


Figure 11. Largest overall weighted topic-share changes; error bars are 95% question-bootstrap intervals. Source: text/topic_prepost.csv and topic_labels.csv. Tool: Matplotlib.

Docker-related topic share rises by 1.28 percentage points, Android by 1.06 and API requests by 0.86. Arrays lose about 1.80 points. Topic shares average normalised document-topic weights using sampling weights. They describe the overall collected corpus; a within-technology denominator can give a different direction. A larger share can coexist with fewer questions in absolute terms.

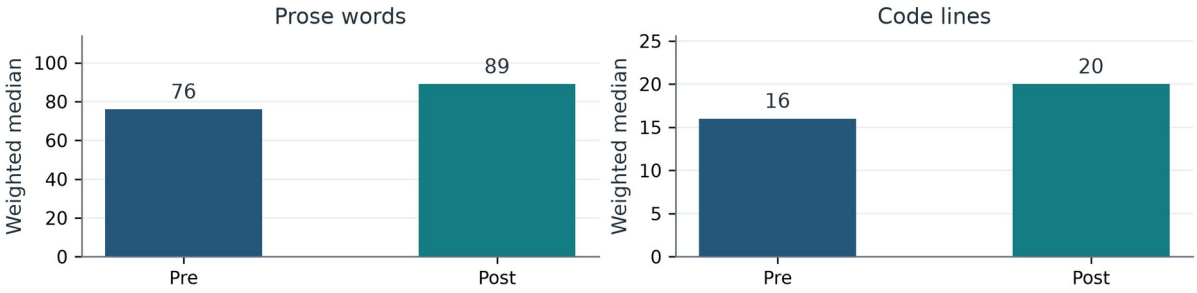


Figure 12. Weighted median changes in question structure. Source: text/complexity_prepost.csv. Tool: Matplotlib.

Prose increases from 76 to 89 words and code from 16 to 20 lines at the weighted median. These measurements describe structure, not verified difficulty. The 1,000 question-level bootstrap resamples do not fully account for dependence among sampled days. Weighted log-odds keyword analysis adds relative vocabulary changes, including later-period terms such as nextjs and vite.

VADER [9] scores original prose with technical-word overrides. Against 200 labels from one human reviewer, accuracy is 36.0% and macro-F1 is 0.344. This establishes weak agreement with that reference, not independent consensus. Tone and frustration-phrase flags remain exploratory and are not used to screen users or allocate support.

6 Comparative Analysis and Business Impact

Table 16. Complementary uses of the two advanced methods

Aspect	Time series	Text mining
Question	How much weekly activity might occur?	What technical themes remain?
Validation	Chronological forecast error and coverage	Coherence, diversity and seed stability
Business use	Capacity scenarios and monitoring	Documentation and expert specialisation
Limit	Structural change and weak original bands	Mixed topics and weak sentiment validity

The 25% relative activity estimate in [1], our fitted slope adjustment and our 26-week projection use different designs and periods. The 52% answer-quality finding in [2] measures a different outcome from our proxy. Our contribution is operational integration, without claiming numerical superiority over either study.

6.1 Review capacity and hypothetical benefit

At approximately 20% review capacity, the revised model finds 58 of 254 proxy-negative cases within 78 reviewed questions (22.83% capture). Same-capacity random selection expects 50.93. For a 1,000-question scenario, assume 200 reviews, five minutes each, INR 500 per reviewer-hour, INR 500 value per additional resolution and INR 1,000 model-operation cost. Shared review costs of about INR 8,333 cancel when comparing equal-capacity policies.

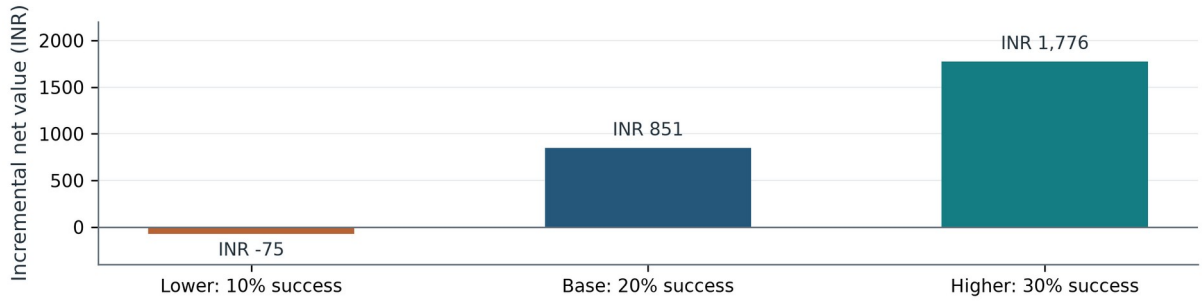


Figure 13. Sensitivity of incremental net value to assumed intervention success. Source: August ranking results and stated cost assumptions. Tool: Matplotlib.

The base case assumes 20% intervention success and produces about 3.70 additional resolutions and INR 851 net value over random routing. At 10% success, net value is about −INR 75; at 30%, it is INR 1,776. These are hypothetical benefits. A controlled pilot must measure actual reviewer time, outcomes and costs. Topic-based documentation and forecast monitoring complement the routing experiment rather than deriving benefit directly from model scores.

7 Interactive Application and Reproducibility

7.1 Evidence exploration and filters

The public Streamlit application [10] integrates eight pages: Overview, Dataset and Classification, Predict a Question, Patterns, Forecasts, Topics, Tone and Structure, and Actions and Scenarios. Pandas filters stored tables, and Plotly redraws interactive charts [11]. Analytical selectors read saved results; they do not retrain the methods.

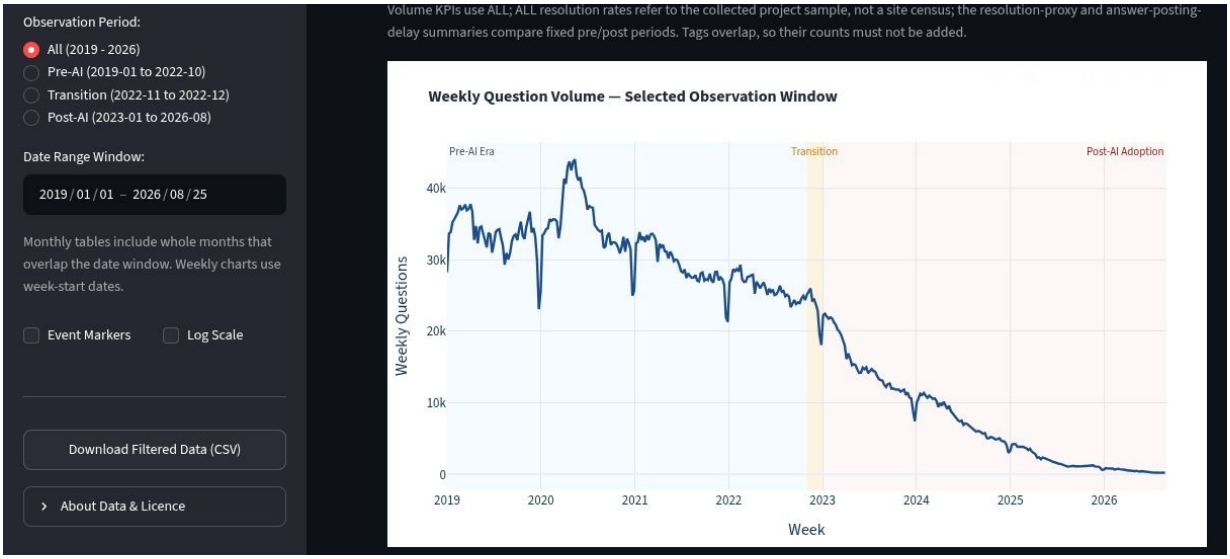


Figure 14. Public dashboard showing the overall activity window. Source: public application, captured 9 November 2026. Tool: browser screenshot.

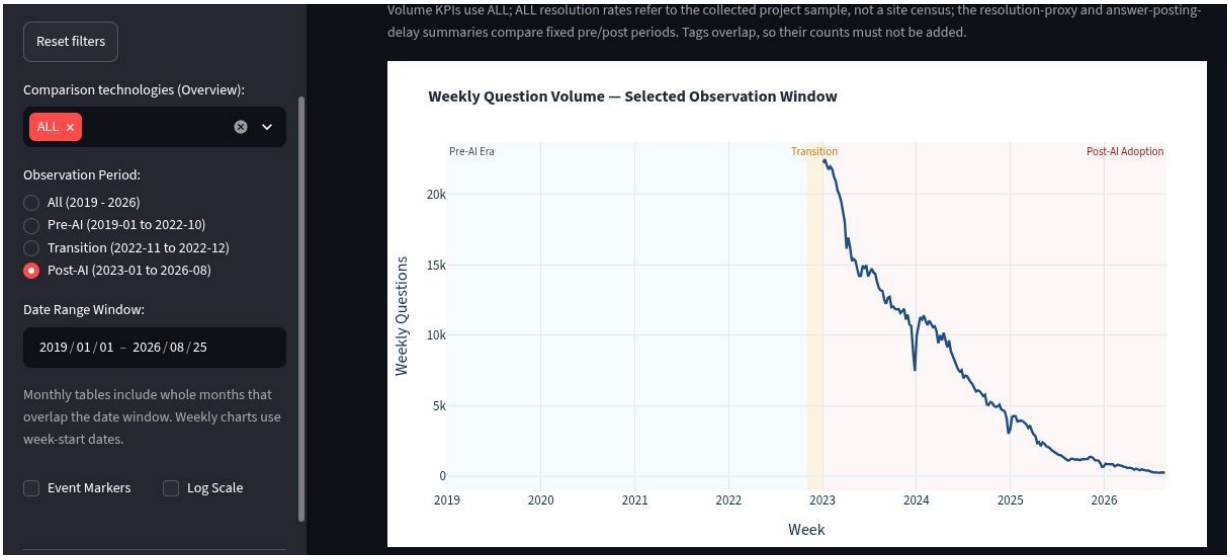


Figure 15. Post-period selection with the corresponding activity chart. Source: same public application and session. Tool: browser screenshot.

These two states demonstrate the changed observation window. Resolution and answer-delay summaries retain their fixed pre/post definition. The interface distinguishes sitewide API totals from weighted question outcomes. The Appendix provides the public address and local launch command.

7 Prediction Explanations and Scenarios

7.2 Dynamic computation

The form runs the fitted feature pipeline, XGBoost and saved calibrator. This example returns 46.8% against threshold 29.7%; tree SHAP explains the raw score. The model demonstrates historical dates through August 2026. Updated history and validation are required for current-date use.



Figure 16. Public prediction and local explanation. Source: application capture, 9 November 2026. Tool: browser screenshot.

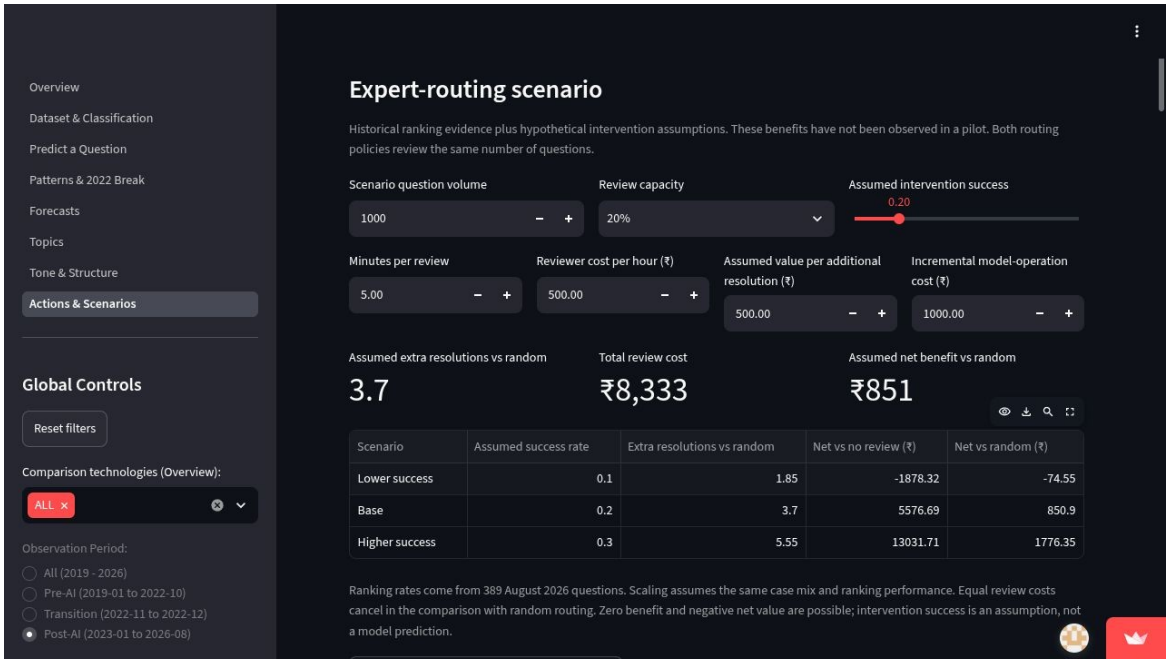


Figure 17. Routing assumptions and equal-capacity random comparison. Source: application capture, 9 November 2026. Tool: browser screenshot.

8 Validation Contributions and Originality

8.1 Objective to evidence traceability

Table 17. Evidence and analytical boundaries

Objective	Evidence	Interpretation / use
Review prioritisation	AP 0.4315 on original locked test; 22.83% August capture at ≈20% capacity	Useful modest ranking; evaluate a controlled pilot
Demand planning	55.9% mean 26-week skill; 52.0% original coverage	Point forecast useful; original bands under-cover
Support themes	NMF 15; diversity 0.735; stability 0.788	Investigate interpretable themes; labels remain provisional
Usable application	Recorded 161-test run; browser evidence; public example reproduced	Software behaviour demonstrated; not a business-impact trial

Dataset checks, chronological evaluation, matched forecasts and preserved outputs provide complementary verification. Snapshot labels and revisions, sampling dependence, overlapping tags and the small August sample constrain generalisation. Single-rater sentiment and automatic topic names do not establish independent human consensus.

8.2 Individual work and tracking

Table 18. Documented contribution areas

Member	Work and evidence
Adhikkesh S K	Classifier development, evaluation and forecasting; contribution_member5.md; PRs 137–139, 172
Abhinav G	API collection, data documentation and dashboard integration; contribution_member1.md; PRs 161–168, 173
Rishiikesh S K	Cleaning, corpus preparation, tone and structure; contribution_member2.md; PRs 129–131, 171
Akshay Krishna P S	EDA, time-series patterns, ITS and synthesis; contribution_member3.md; PRs 146, 147, 148, 150
Nishanth K	Feature engineering, topic modelling and keyword comparison; contribution_member4.md; PR 140

The Appendix provides the board snapshot and link. Card status records tracking; analytical validity is assessed separately. Commits, PRs and contribution files retain work history.

8.3 AI use and originality

We used Claude, OpenAI Codex and Antigravity for coding, debugging, explanations, workflow illustration and writing. We checked reported values against saved outputs, compared figures with source tables, and verified behaviour through recorded tests and browser observations. We distinguished observations, predictions and assumed benefits. Assisted annotations remain separate from human reference labels. Sources are cited; the team accepts responsibility for the submitted analysis.

9 Conclusion Recommendations and Roadmap

The project connects a directly collected and validated dataset with chronological question classification, time-series forecasting and text analysis. The results provide a practical way to inspect changing support demand. Classification captures useful but modest ranking information, forecasts describe continued contraction with uncertain bands, and text mining identifies shifts in the composition of the collected questions.

The evidence matrix in Section 8 distinguishes implementation results from operational outcomes. The strongest immediate use is a monitored decision-support pilot. Neither the observational time break nor the later forecast identifies the causal effect of any one technology. The resolution label remains an accepted-answer posting proxy rather than a direct measurement of satisfaction or correctness.

Table 19. Prioritised recommendations and implementation roadmap

Timing / owner	Action	Success check
Immediate: forum manager and experts	Compare model-ranked and random review queues at the same review budget	Queue age, proxy outcomes, minutes per review and incremental cost
Short term: documentation lead	Choose an interpretable recurring theme and test a guide or question template	Guide use, repeat-question rate and review outcomes against a comparison group
Medium term: analyst	Monitor forecast errors and reassess interval coverage on new outcomes	Rolling error, empirical coverage and alert burden
Medium term: model owner	Monitor probability calibration and performance across technologies and account groups	Brier score, capacity capture and subgroup sample sizes
Longer term: data owner and modeller	Refresh history, reconstruct revisions where available and evaluate a genuinely later period	Consistent measurements and prospective evidence before wider use

9.1 Future scope

Further work should improve the measurements before adding model complexity. Original revision histories and acceptance-event timestamps would strengthen posting-time reconstruction. Broader technology coverage and sampling-aware uncertainty would improve generalisation. Independent topic interpretation and a multi-rater sentiment reference would strengthen semantic validation. A prospective routing pilot is needed to establish whether expert attention changes outcomes and whether any benefit exceeds its cost.

References

[1] R. Maria del Rio-Chanona, Nadzeya Laurentsyeva, and Johannes Wachs. 2024. Large language models reduce public knowledge sharing on online Q&A platforms. PNAS Nexus 3, 9, pgae400. https://doi.org/10.1093/pnasnexus/pgae400

[2] Samia Kabir, David N. Udo-Imeh, Bonan Kou, and Tianyi Zhang. 2024. Is Stack Overflow Obsolete? Proceedings of the CHI Conference on Human Factors in Computing Systems, Article 935, 1–17. https://doi.org/10.1145/3613904.3642596

[3] Stack Exchange. n.d. Stack Exchange API v2.3 documentation and source question records. Accessed 9 November 2026. https://api.stackexchange.com/docs . Source attribution and content licence: https://stackoverflow.com/help/licensing

[4] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. Proceedings of KDD, 785–794. https://doi.org/10.1145/2939672.2939785

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[13] Project Team 7. 2026. Stack Overflow in the AI Era: code, data documentation and analytical evidence. Main implementation revision d447635. https://github.com/Abhinav-Gopalakrishnan/stackoverflow-ai-era

Analytical figures were generated with Matplotlib [12]; model selection and transformations use scikit-learn [5]. Project-specific configurations, saved results and source files are retained in [13].

Appendix Evidence and Reproduction

A Project board and contribution history

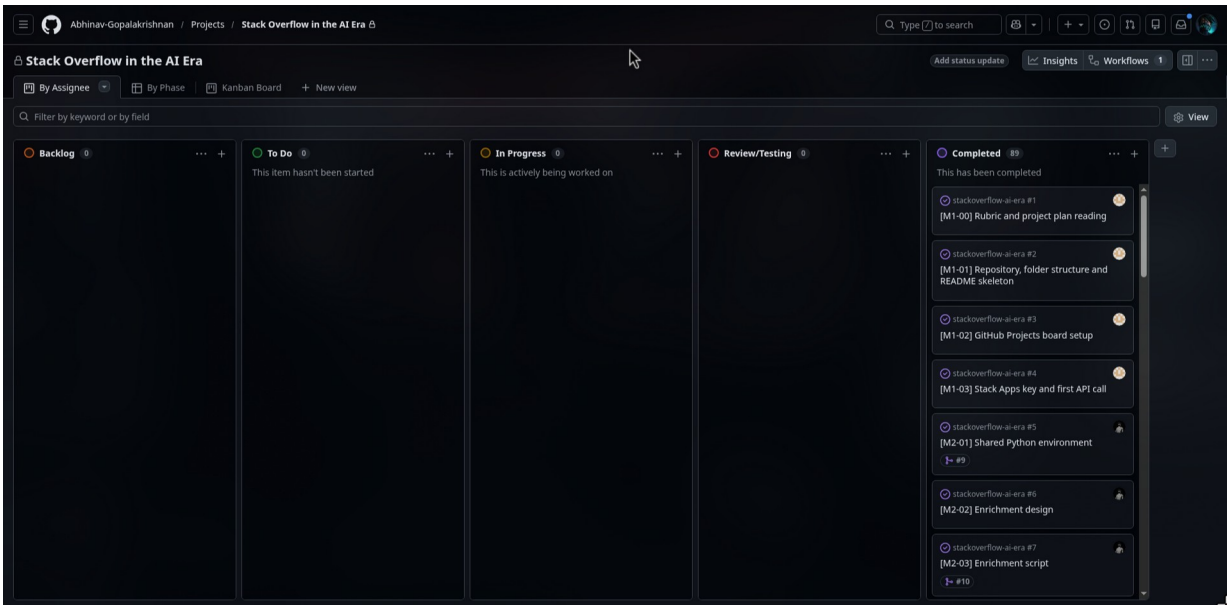


Figure 18. Supplied GitHub Projects snapshot showing the five workflow stages and 89 cards marked Completed. Source: team-provided board capture. Tool: browser screenshot.

Board: https://github.com/users/Abhinav-Gopalakrishnan/projects/3 . Contribution files are under reports/review1/ and reports/review2/. The snapshot documents the displayed tracking state; method validity is addressed in Section 8.

B Dataset and analytical evidence

Dictionary: docs/data_dictionary.md. Validation: reports/review2/dataset/dataset_validation.json. Full dataset: questions_clean_v1.parquet, 316,393,537 bytes, distributed separately from Git. SHA-256: dd4a5d96df1cc93e1ccbad625bc51f01c65a96a32bce452469f1e62662dfaeaf.

Original classifier: reports/review1/m5/. Revised classifier: reports/final_validation/. Time series: reports/review2/ts/. Text mining: reports/review2/text/. This report folder preserves figure source CSVs, PNG/SVG exports and _build/generate_figures.py. Final notebooks: notebooks/review1_main.ipynb and notebooks/review2_main.ipynb; release details: docs/final_implementation.md.

C Application and reproduction

Public application: https://stackoverflow-ai-era.streamlit.app/ . Verified runtime: Python 3.13.13. From the repository root:

```
python3.13 -m venv .venv
.venv/bin/python -m pip install -r dashboard/requirements.txt
.venv/bin/python -m streamlit run dashboard/app.py
```

Open http://localhost:8501. The committed model and aggregate exports support the application. Full analysis needs the separately supplied data and language resources. Install requirements-dev.txt for the analysis environment; run python -m src.ts_patterns, python -m src.ts_forecast, python -m src.text_prep and python -m src.topics to reproduce the advanced methods. Released model rebuilding requires an explicit --replace option.